
ENGINEERING PROJECT PREPERATION - 42908

Literature Review

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A Comparative Life Cycle Cost Analysis of Sustainable and Conventional Buildings

Literature survey

The Life Cycle Costing (LCC) is widely used economic evaluation method that calculates the total cost of a building throughout its entire life span. This includes prior construction costs, operational expenditure, maintenance, and disposal costs. Research indicates that operational cost is about 60–80% of the total life cycle costs, which highlight their importance in cost assessments.

Sustainable buildings are constructed mainly to minimize environmental impact and reduce resource use. But, studies convey that their initial construction costs are generally 5–15% higher than those of conventional buildings.

Despite of expensive prior construction cost, multiple studies determine that sustainable buildings are:

- 20–40% Efficient in energy consumption
- 30–50% Efficient in water usage
- 10–25% less maintenance costs

Over a typical lifespan of 30–50 years, these savings convert into 10–20% of less total life cycle costs compared to conventional buildings.

However, results vary based on assumptions such as discount rates (commonly 4–7%) and annual energy price escalation (2–5%), underscoring the importance of standardized analysis methods.

Over **75%** of new homes in NSW now meet 7-star energy ratings, showing adaptability of energy-efficient design. Upgrading energy efficiency standards increases construction cost by only about AUD \$4,300 per home, which is relatively small compared to total construction cost

Significance:

1.1 Cost Optimization

Life Cycle Costing (LCC) mainly focused on the identification of long-term cost savings, For which studies indicates up to 20% redemption in total ownership costs when sustainable designs are applied.

1.2 Sustainability Integration

as buildings represent around 40% of global energy consumption, incorporating sustainable design is important for the effective redemption of environmental impact.

1.3 Industry Relevance

In Australia, Especially construction costs have increased by approximately 6–10% annually, making effective cost planning important for project efficiency.

1.4 Policy Support

Green buildings can reduce greenhouse gas emissions by 30–50%, which contribute significantly to national sustainability goals.

Objectives:

- To compare the life cycle costs (LCC) of sustainable and conventional buildings.
- To evaluate operational cost savings in energy and water consumption.
- To quantify long-term financial benefits using Life Cycle Costing (LCC).
- To analyze the sensitivity of results to key financial parameters such as discount rates and energy price escalation

Present Study:

This study adapts a comparative case study approach, utilizing secondary data sourced from cost reports and sustainability regulations.

The analysis is based on the following assumptions:

- Building lifespan: 40 years
- Discount rate: 5%
- Energy cost escalation: 3% annually

Life Cycle Cost (LCC) calculations will compare:

- Initial construction cost
- Annual operational cost
- Maintenance and replacement cost

Cost Component	Sustainable Building	Conventional Building
Initial Cost	+10% higher	Base value
Energy Cost	25–40% lower	Higher
Water Cost	30–50% lower	Higher
Maintenance	10–20% lower	Moderate
Total LCC	10–20% lower (long-term)	Higher

Literature Summary Table:

Author	Year	Description	Conclusion
Zuo et al.	2017	Analysed cost and energy performance of green buildings	25–30% reduction in energy cost observed
Cabeza et al.	2014	Evaluated LCC methodology in construction	LCC improves decision accuracy by ~15–20%
GBCA Reports	2020	Green Star certified buildings analysis	Sustainable buildings reduce emissions by up to 50%

Identify Research Gap:

- Limited Sydney local specific data although Australia’s unique cost structure.
- Lack of studies, lead to apply consistent discount rates within the 4–7% range.
- Few studies integrating real Quantity Surveyor (QS) cost data with sustainability metrics.
- Limited statistical comparisons based on actual cost breakdowns.
- Uncertainty in long-term predictions caused by fluctuating energy prices.

Industrial Relevance:

- Construction Industry - Potential cost savings of 10–20% over the building lifecycle.
- Real Estate - Higher asset value with lower operating costs.
- Government Projects - Long-term savings in infrastructure investment.

- Sustainability Sector - Supports achievement of carbon reduction targets.

Challenges:

- Difficulty in precise prediction of costs over a 30-50 year period.
- fluctuation in energy price growth rates, typically ranging from 2–5% annually
- Uncertainty in collecting reliable and consistent long-term data.
- Sensitivity of Life Cycle Costing (LCC) results to changes in discount rates.

Factors Affecting Life Cycle Cost:

- Discount rate: 4–7%
- Energy price escalation: 2–5% annually
- Building lifespan: 30–50 years
- Material durability
- Maintenance cycles

Proposed Research Questions:

1. To what extent are sustainable buildings more cost-effective than conventional buildings over their life cycle in Sydney?
2. How much operational cost reduction (in %) contributes to total LCC savings?
3. How sensitive are LCC results to discount rate and energy price variations?.
4. What is the percentage difference in total life cycle cost between sustainable and conventional buildings?

Conclusion:

Sustainable buildings involve higher initial costs, typically 5–15% more than conventional buildings. Moreover, they achieve substantial operational savings ranging from 20–40%, which leads to a reduction of 10–20% in total life cycle costs over time.

Lastly, due to fluctuation in underlying assumptions and limited localized data, further research is necessary to validate these findings specifically within the Sydney.

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